

EXPO BASE

Universal application foundation

A production-oriented Expo reference app proving adaptive composition, semantic geometry, portable services, and consistent interaction behavior across compact through wide layouts.

- Adaptive
- Portable

 System lab

 Stress test

Adaptive foundation

One hierarchy spans compact through wide layouts while surfaces express primary, supporting, and contextual information without feature-owned geometry.

- Compact

Adaptive regimes

4

Compact · medium · expanded · wide

Target platforms

3

iOS · Android · web

Minimum interaction target



Home



Build



Data



Patterns



System

Form interaction acceptance surface

Keyboard-aware scrolling, visible labels, deterministic focus, validation summary, server-error mapping, and unsaved-change protection remain outside the visual component package.

 Back

IDENTITY

Profile information

Related identity fields share one responsive section while each field keeps its own validation and keyboard contract.

Required

Full name

Required

Alex Morgan

Email address

Required

alex@example.com

SECURITY

Account security

Security inputs remain visually grouped without coupling the forms package to a state library.



Home



Build



Data



Patterns



System

Navigation acceptance surface

The visual navigation system is route-library agnostic. Expo Router integration is isolated behind @expo-base/navigation-router.

← Back

Primary destination model

Compact layouts use a maximum of five core destinations. Expanded layouts promote the same information architecture into a persistent sidebar instead of inventing a separate route model.



Home



Cards



Rewards



Activity



Profile

Selected: home

Tabs

Overview

Rewards

Activity

Be



Selected tab: overview. The tab row owns its



Home



Build



Data



Patterns



System

Overlay Manager

acceptance surface

Anchored surfaces measure their real window position, flip when necessary, clamp to safe viewport bounds, and share exclusive lifecycle ownership.

← Back

Anchored popover

The trigger is measured with `measureInWindow` after render. The panel remains invisible until both anchor and content geometry are known, preventing a random first-frame jump.

Open action menu ▾

Collision-aware

Dialog / alert / sheet hierarchy

Open dialog

Open destructive alert

Open bottom sheet

Open long sheet

Show toast



Home



Build



Data



Patterns



System

Virtualized list acceptance surface

This route renders 250 synthetic rows through one screen-owned FlatList. Product code never owns raw list virtualization or nested vertical scrolling.

 Back

Production list states

Loading, empty, and live-data ownership stay inside the shared list contract.

Stateful

Live data

Loading state

Empty state

Small static groups

StaticList remains the bounded choice for small groups embedded inside another scroll owner.

≤40 items

Small groups embedded inside another scroll owner use StaticList. Above the contract threshold, development warns and the screen should move to ListScreen.

Profile settings


Home


Build


Data


Patterns


System

GATE 09 / DATA & CONTENT

Adaptive data workspace

Search, filter, inspect, and paginate one semantic dataset while the shared system owns responsive composition, density, formatting, and state hierarchy.

Responsive data

Search + filters

Total annual fees

\$1,215.00

7 cards tracked

Estimated reward value

\$4,940.00

Synthetic reference data

Average utilization

12.3%

Central numeric formatting

Portfolio cards

The toolbar, result state, adaptive table, and pagination behave as one reusable data workspace rather than feature-owned layout fragments.

7 records

🔍 Search card, issuer, or status



Home



Build



Data



Patterns



System

GATE 10 / VISUALIZATION

Tokenized charts with bounded geometry

Core charts share one series palette, measured chart frame, scale/path solvers, accessible summaries, and tap selection. Product screens never calculate SVG paths or import a chart engine directly.

Portfolio value

Area treatment, semantic accent series, quiet grid, tap-selection zones.

Interactive



Label	Value
Jan	6200
Feb	6480
Mar	6310
Apr	6840
May	7010



Home



Build



Data



Patterns



System

The unhappy path is part of the design system

Loading, stale content, empty results, errors, offline behavior, permissions, reconnect flows, and persistent alerts use one semantic language instead of feature-specific improvisation.

Persistent semantic feedback



New recommendation available

Your current spending pattern may qualify for a stronger dining return.

Review



Bonus completed

Required spend was reached nine days early.



Annual fee in 12 days

Review retention value before the renewal posts.



Connection failed

The latest refresh could not be completed.



Inline messages are reserved for local context such as form or section-level status.



Home



Build



Data



Patterns



System

GOLDEN PHASE 3

Server-state lifecycle laboratory

One scoped query owner coordinates identity, cache, stale refresh, cancellation, mutation invalidation, and optimistic rollback across consumers.

 Back

Shared query identity

Three consumers use one deterministic key; the service receives one request.

Service requests: 1

Consumer A: content · 3

Consumer B: content · 3

Recompose runtime provider

All tasks

Platform tasks

Stale-safe presentation

Refresh errors retain content and map into the shared degraded feedback anatomy.



Home



Build



Data



Patterns



System

GATE 12 / MOTION + ACCESSIBILITY

Motion is optional; meaning is not

Animation and haptics are progressive enhancement. Reduced-motion policy, live announcements, semantic grouping, and screen-reader-only content are centralized so interaction remains understandable without movement or vibration.

Reduced motion

System preference detected: Reduced motion enabled.

Hide reveal

Purposeful transition

With reduced motion enabled, Reanimated reaches the same final state without requiring the movement.

Haptic intent

This lab uses a deterministic fake adapter. Production haptics are optional and silently degrade when hardware/browser support is unavailable.

Selection

Confirm

Warning



Home



Build



Data



Patterns



System

Dashboard

Primary overview pattern for consumer or operational products.

- 1 / 15
- Next pattern
- Home

Tracked value

\$8,420

+14.2% YTD

Annual fees

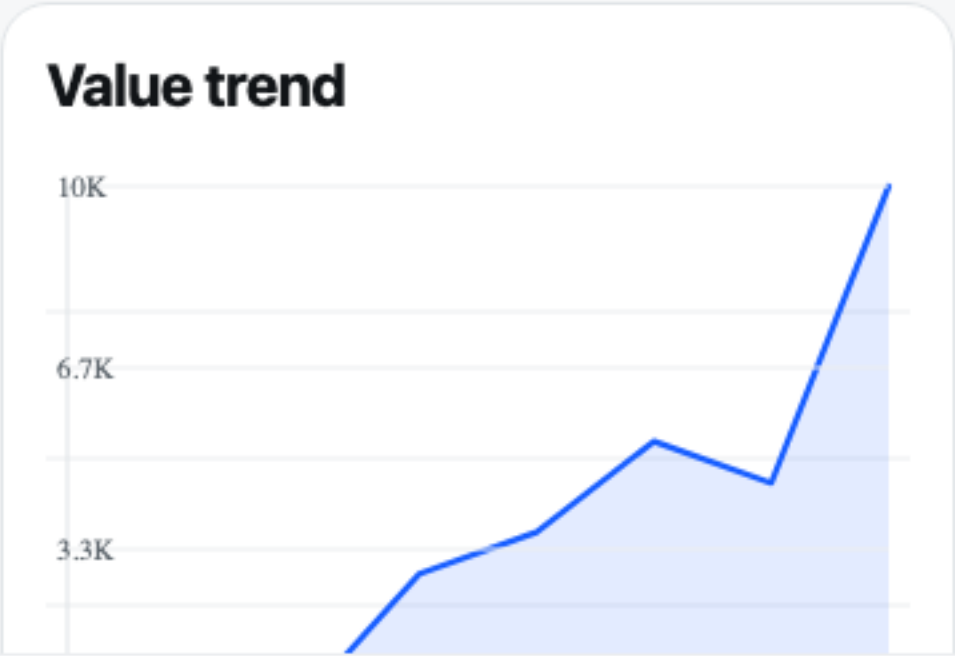
\$1,215

6 active products

Upcoming value

\$1,090

3 benefits



- Home
- Build
- Data
- Patterns
- System

WORKFLOW LAB

Activity

Offline is honest: retained content remains visible, but write synchronization is not implied.



Commands

Home



Offline

Showing available information. Changes may not be current.

Recent activity

This retained sample content represents a stale server-state result. Explicit refresh remains a product action after reconnect.

Refresh when online



Home



Build



Data



Patterns



System

GATE 14 / COMPONENT LAB

System acceptance laboratory

A single route exercising public components in realistic combinations so theme, density, interaction and responsive regressions are visible before product migration.

Stress cases →

Home

Surface hierarchy

SURFACE

Default workspace surface

Stable bordered surface for most grouped application content.

SUBTLE

Contextual surface

Lower-emphasis grouping for inspectors, supporting context, and quiet states.

ELEVATED

Prominent surface

Raised treatment for authentication. hero tasks.



Home



Build



Data



Patterns



System

GOLDEN PLUS

Abundant reusable depth

A deterministic living specification for secondary UI families that should never be rebuilt in product routes.

No external runtime dependency

Identity and status

Image fallback, overlap, aggregate naming, and non-color status semantics remain centrally owned.

AK

ML

SP

+2



Service healthy

All scheduled checks completed.



Review required

One policy needs an owner.

Workspace identifier

workspace_01J8QZ6VBH79M5X2



Sensitive reference

.....9831

Reveal



```
{
  "density": "compact",
  "locale": "en-US"
}
```



Home



Build



Data



Patterns



System

GATE 14 / TORTURE MATRIX

Pathological-content acceptance

This screen intentionally feeds the design system long labels, large values, multilingual content, stale-data conditions and constrained action groups. It should wrap or adapt without overlap, clipping or page-level horizontal scrolling.

Component lab

Extreme values

Very large currency

\$9,999,999,999.99
+999.99%

Negative balance

-\$8,420,293.18
Requires attention

Large points balance

123,456,789 pts
Across 18 programs

Long and multilingual text

LONG ENGLISH



Home



Build



Data



Patterns



System

Service + auth runtime acceptance surface

Feature code consumes portable service contracts and centralized auth state. Backend adapters remain hidden behind runtime boundaries.

[← Back](#)

Authenticated

Reference User

Adapters ready

Auth runtime

reference@example.com

Image provider

reference://asset

Storage

Injected KeyValueStorageAdapter

Analytics

Injected AnalyticsAdapter

Sign out through auth runtime

Verify storage adapter



Home



Build



Data



Patterns



System

RUNTIME CAPABILITY LAB

Optional boundaries, deterministic demo

This reference route demonstrates selection and normalized outcomes through fake adapters only. Production apps select real Expo implementations in root composition.

Storage and sharing

Save preference

Store secret

Share value

Signals and permissions

Read connectivity

Local unlock

Notification token

Release boundary

Check update

Back to system

All interactions below use deterministic memory fixtures; no hardware, credentials, or backend are involved



Home



Build



Data



Patterns



System

Authentication session acceptance

This route is itself protected. Signing out must remove it from navigation history and expose only signed-out routes.

Resolution
signed-in

Access
granted

Action
idle

User
reference@example.com

Sign out and test guard



Home



Build



Data



Patterns



System

Authorization capability acceptance

Authenticated identity and product capability are separate. Loading/error states fail closed and backend APIs must still enforce authorization.

ready

Capabilities

reports.view, rewards.optimize, settings.manage

reports.view

Granted

billing.manage

missing_all

Reports capability content is visible.

Open capability-protected route

Refresh capabilities



Home



Build



Data



Patterns



System

ANALYTICS / OVERVIEW

A calm view of growth

A reusable analytics composition connecting trend, comparison, distribution, and accountable detail without turning the page into a chart gallery.

Active accounts

18,420

↑ 12.8% vs prior period

Healthy acquisition



Net expansion

\$104.3k

↑ 8.4% vs prior period

Above target

Retention floor

91.4%

↓ 1.2 pts below target

Watch enterprise cohort

Monthly performance

Current period against the previous period and operating target

↑ Current period is ahead of target in four of six months.

100



Home



Build



Data



Patterns



System

FINANCE / OPERATIONS

Value, contribution, and control

A neutral financial-operations composition demonstrating signed change and reference levels without embedding advice or financial business rules.

Managed value

\$1.04m

↑ 6.2% above prior period

Reference value

Net movement

+\$203k

↑ Expansion offsets churn

Semantic tone supplied by the product

Operating target

\$1.00m

↑ 4.3% above target

Target remains visible

Period comparison

Current and previous values share one ordered x-axis.



Home



Build



Data



Patterns



System

MONITORING / OPERATIONS

Know what needs attention

A dense operational composition connecting health, thresholds, anomalies, and recent events with deterministic sample data.

- All systems operational
- 2 services under watch

Availability

99.98%

↑ 0.04 pts above target

Last 24 hours

P95 latency

182ms

↓ 18ms above target

Checkout path

Open incidents

2

↑ 1 new in the last hour

Acknowledgement required

Service health trend

The selected threshold is visible in the surrounding operationa

100

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GOLDEN PATTERN / FEED-LIST

Activity feed

Dense chronological or entity collections with optional controls.

2 / 15

Next pattern

Home

All

Rewards

Spend



Venture X

Capital One · Updated 4 min ago

Active



Gold Card

American Express · Review annual fee

Review



Sapphire Preferred

Chase · Travel category

Search results

Results remain primary; compact layouts place filters before the result set.

3 / 15

Next pattern

Home

24 results for "travel"

Filters

Issuer

Annual fee

Status



Venture X

Capital One · Updated 4 min ago

Active



Gold Card

American Express · Review annual fee

Review



Sapphire Preferred

Chase · Travel category



Home



Build



Data



Patterns



System

Data workspace

Wide operational surface with optional inspector instead of a squeezed mobile table.

Filter

Export

Card	Venture X
Issuer	Capital One
Value	\$1,220.00

Card	Gold Card
Issuer	American Express
Value	\$980.00

Card	Sapphire Preferred
Issuer	Chase
Value	\$690.00

Scope
All cards
Updated
4 min ago

GOLDEN PATTERN / DETAIL

Venture X

Entity detail pattern with hero summary and contextual aside.

5 / 15

Next pattern

Home

Tracked value

\$8,420

+14.2% YTD

Annual fees

\$1,215

6 active products

Upcoming value

\$1,090

3 benefits

Issuer

Capital One

Annual fee

\$395

Opened

May 2026



Home



Build



Data



Patterns



System

GOLDEN PATTERN / MASTER-DETAIL

Cards

Selection and detail coexist only when the layout capability supports it.

- 6 / 15
- Next pattern
- Home



Venture X

Capital One · Updated 4 min ago

Active



Gold Card

American Express · Review annual fee

Review



Sapphire Preferred

Chase · Travel category

Edit setup

Single-purpose create/edit flow with centralized form action hierarchy.

Basics

Labels remain visible and validation behavior is system-owned.

Nickname

Travel setup

Institution

Capital One

Save changes

Cancel

GOLDEN PATTERN / WIZARD

Connect account

Progressive multi-step workflows keep progress and actions in stable locations.

8 / 15

Next pattern

Home

Step 2 of 4

Confirm institution

Review the selected provider before continuing.

Institution

Example Bank

Connection

Secure OAuth

Continue

Back



Home



Build



Data



Patterns



System

Settings

Settings become a master/detail workspace on larger layouts and a safe stack on compact layouts.

- 9 / 15
- Next pattern
- Home

General

Notifications

Privacy

General

Appearance

System theme

Density

Comfortable

Language

English

Profile

Identity summary followed by independent profile sections.

10 / 15

Next pattern

Home

Alex Kim

alex@example.com

Verified

Personal information

Name

Alex Kim

Timezone

Central

Security

Passkey

Enabled

Recovery

Ready



Home



Build



Data



Patterns



System

EXPO BASE

Welcome back

Authentication stays focused and deliberately narrow on large screens.

Email

alex@example.com

Password

.....

Sign in

11 / 15

Next pattern

Home



Home



Build



Data



Patterns



System

Analytics

Metrics and visualizations scale independently while insight context moves to an aside.

12 / 15

Next pattern

Home

Tracked value

\$8,420

+14.2% YTD

Annual fees

\$1,215

6 active products

Upcoming value

\$1,090

3 benefits

Value over time



Home



Build



Data



Patterns



System

GOLDEN PATTERN / EMPTY-START

Cards

New-user and zero-data experiences preserve hierarchy rather than showing a blank page.

13 / 15

Next pattern

Home



Add your first card

Start with one account. The platform will build the rest of the experience around real content.

Add card

Learn more



Home



Build



Data



Patterns



System

GOLDEN PATTERN / FULL-SCREEN-WORKFLOW

Review changes

Focused workflows keep one dominant task and stable exit/continue actions.

14 / 15

Next pattern

Home

 3 changes ready

Nothing is submitted until you confirm.

Nickname

Travel setup

Issuer

Capital One

Status

Active

Confirm changes

Back

Overlay workflow

Page context stays intact while a shared dialog or sheet owns transient decisions.

15 / 15

Next pattern

Home

Context remains visible

Open the system-owned confirmation overlay without feature-level positioning or z-index.

Open confirmation



Home



Build



Data



Patterns



System

WORKFLOW LAB

Import complete

Completion owns the handoff hierarchy, not product geometry.



Commands

Home



Completed

Your file is ready for product-specific processing.

Products supply the destination, copy, and domain result.

View imports

Start another



Home



Build



Data



Patterns



System

WORKFLOW LAB

Allow document access

The capability package owns availability and permission state; product copy owns the reason.



Commands

Home



Permission required

Document selection is optional. You can continue without importing a file.

Choose a document

Not now



Home



Build



Data



Patterns



System

Virtualized list acceptance surface

This route renders 250 synthetic rows through one screen-owned FlatList. Product code never owns raw list virtualization or nested vertical scrolling.

 Back

Production list states

Loading, empty, and live-data ownership stay inside the shared list contract.

Stateful

Live data

Loading state

Empty state

Small static groups

StaticList remains the bounded choice for small groups embedded inside another scroll owner.

≤40 items

Small groups embedded inside another scroll owner use StaticList. Above the contract threshold, development warns and the screen should move to ListScreen.

Profile settings


Home


Build


Data


Patterns


System

GATE 08 / LISTS

Virtualized list acceptance surface

This route renders 250 synthetic rows through one screen-owned FlatList. Product code never owns raw list virtualization or nested vertical scrolling.

← Back

Production list states

Loading, empty, and live-data ownership stay inside the shared list contract.

Stateful

Live data

Loading state

Empty state

Small static groups

StaticList remains the bounded choice for small groups embedded inside another scroll owner.

≤40 items

Small groups embedded inside another scroll owner use StaticList. Above the contract threshold, development warns and the screen should move to ListScreen.

Profile settings



Home



Build



Data



Patterns



System

GATE 06 / NAVIGATION

Navigation acceptance surface

The visual navigation system is route-library agnostic. Expo Router integration is isolated behind @expo-base/navigation-router.

← Back

Primary destination model

Compact layouts use a maximum of five core destinations. Expanded layouts promote the same information architecture into a persistent sidebar instead of inventing a separate route model.



Home



Cards



Rewards



Activity



Profile

Selected: home

Tabs

Activity

Benefits with a deliberately long label

Selected tab: benefits The tab row owns its



Home



Build



Data

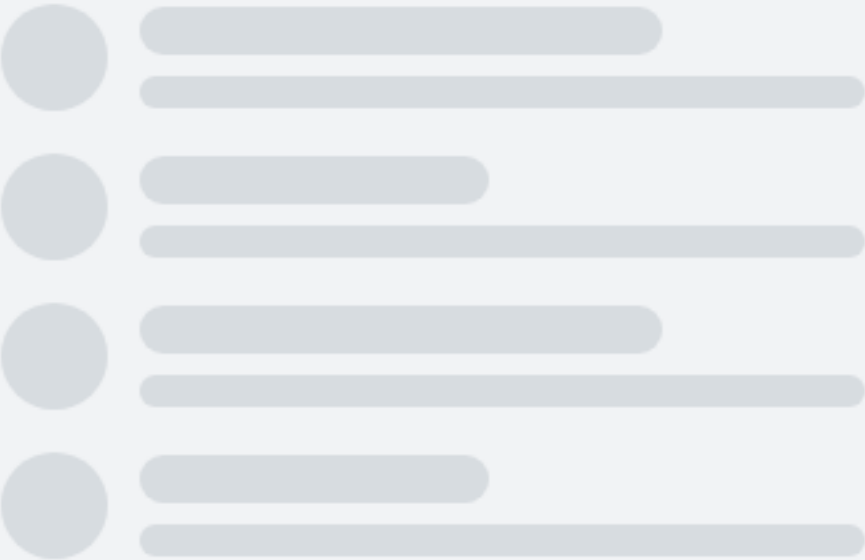


Patterns

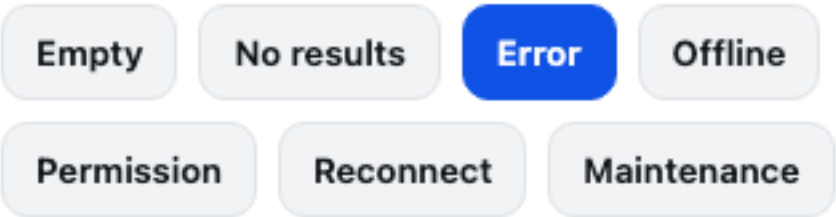


System

Loading skeleton



State playground



Something went wrong

Try again. If the problem continues, the service may be temporarily unavailable.





[Redacted text]



[Redacted text]



[Redacted text]



[Redacted text]

State playground

Empty

No results

Error

Offline

Permission

Reconnect

Maintenance



Permission required

Grant the required access to continue this workflow.

Try again

Stale-content preservation



Home



Build



Data



Patterns



System



[Redacted text]



[Redacted text]



[Redacted text]



[Redacted text]

State playground

Empty

No results

Error

Offline

Permission

Reconnect

Maintenance



Temporarily unavailable

This area is undergoing maintenance.
Try again later.

Try again

Stale-content preservation



Home



Build



Data



Patterns



System

GATE 09 / DATA & CONTENT

Adaptive data workspace

Search, filter, inspect, and paginate one semantic dataset while the shared system owns responsive composition, density, formatting, and state hierarchy.

Responsive data

Search + filters

Total annual fees

\$1,215.00

7 cards tracked

Estimated reward value

\$4,940.00

Synthetic reference data

Average utilization

12.3%

Central numeric formatting

Portfolio cards

The toolbar, result state, adaptive table, and pagination behave as one reusable data workspace rather than feature-owned layout fragments.

0 records

🔍

no-such-card

✕

Stale-state presentation

Refresh errors retain content and map into the shared degraded feedback anatomy.



Update failed

Showing previously loaded data because the latest refresh failed.

Try again

Publish Golden Catalog

platform · Complete

Adopt shared query lifecycle

product · Open

Review stale-data behavior

platform · Open

Refresh tasks

Fail next refresh

Mutation bookkeeping

The product chooses the optimistic update; the shared owner snapshots, rolls back, invalidates, and single-flights.

Mutation: idle

Mutation requests: 0



Home



Build



Data



Patterns



System

Email address: This email address is already in use.

IDENTITY

Profile information

Related identity fields share one responsive section while each field keeps its own validation and keyboard contract.

Required

Full name

Required

Alex Morgan

Email address

Required

alex@example.com

 This email address is already in use.

SECURITY

Account security

Security inputs remain visually grouped without coupling the forms package to a state library.

10+ characters

Password

Required



At least 10 characters.



Home



Build



Data



Patterns



System

GATE 07 / OVERLAYS

Overlay Manager

acceptance surface

Anchored surfaces measure their real window position, flip when necessary, clamp to safe viewport bounds, and share exclusive lifecycle ownership.

← Back

Anchored popover

The trigger is measured with measureWindow

Remove this configuration?

Alert dialogs require explicit action and do not dismiss on backdrop.

Cancel

Remove

Dialog / alert / sheet hierarchy

Open dialog

Open destructive alert

Open bottom sheet

Open long sheet

Show toast



Home



Build



Data



Patterns



System

Capability-protected route

This route exists only while the authenticated identity has settings.manage.

Back



Home



Build



Data



Patterns



System

Semantic status

Feedback colors remain semantic and theme-safe rather than feature-owned.

Approved

Review

Blocked

Info

Runtime controls

Switch theme, density, locale direction, and motion policy live. Geometry remains token-owned; density never reduces interaction safety.

Theme

System

Light

Dark

Density

Comfortable

Compact

Locale stress

English

Pseudo LTR

Pseudo RTL

Motion

System motion

Reduced motion

Active: Dark theme · Comfortable density · en-US · LTR · system motion

Locale mode: Product locale · Direction: LTR

Adaptive split



Home



Build



Data



Patterns



System